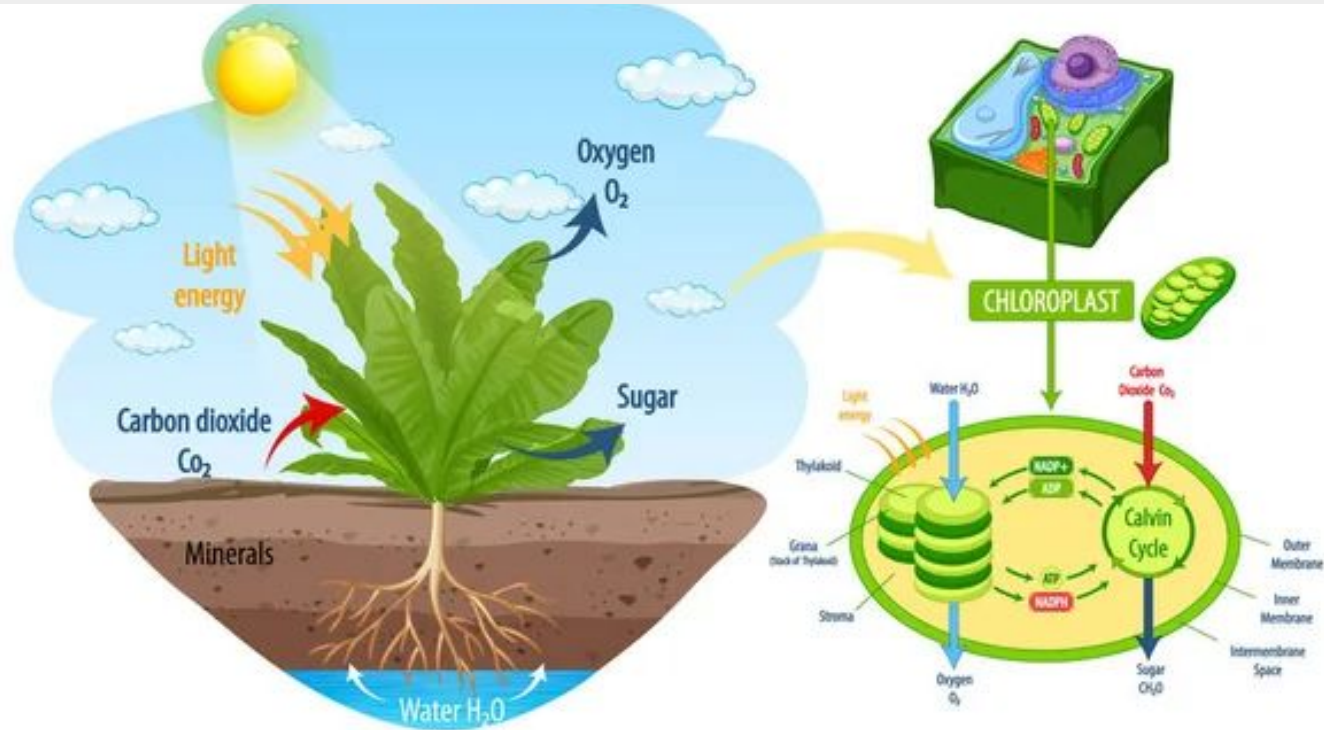


INTRODUCTION TO BIOLOGY

Plants for People and Environment

Lecture 06

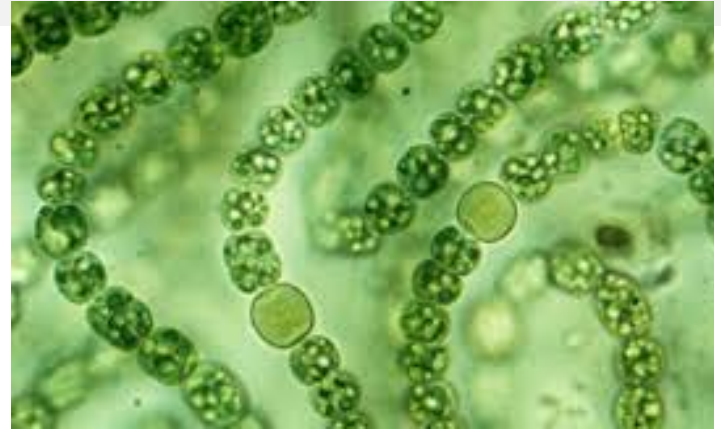


Primary Producers

Green plants and certain types of bacteria and algae are the primary producers as they can produce usable energy for the rest of the living organisms on the planet. Energy from the sun is used to make glucose which all the other forms of life take in and "burn" for energy.

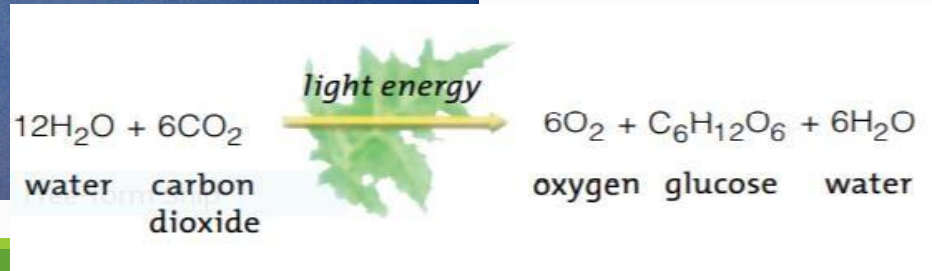
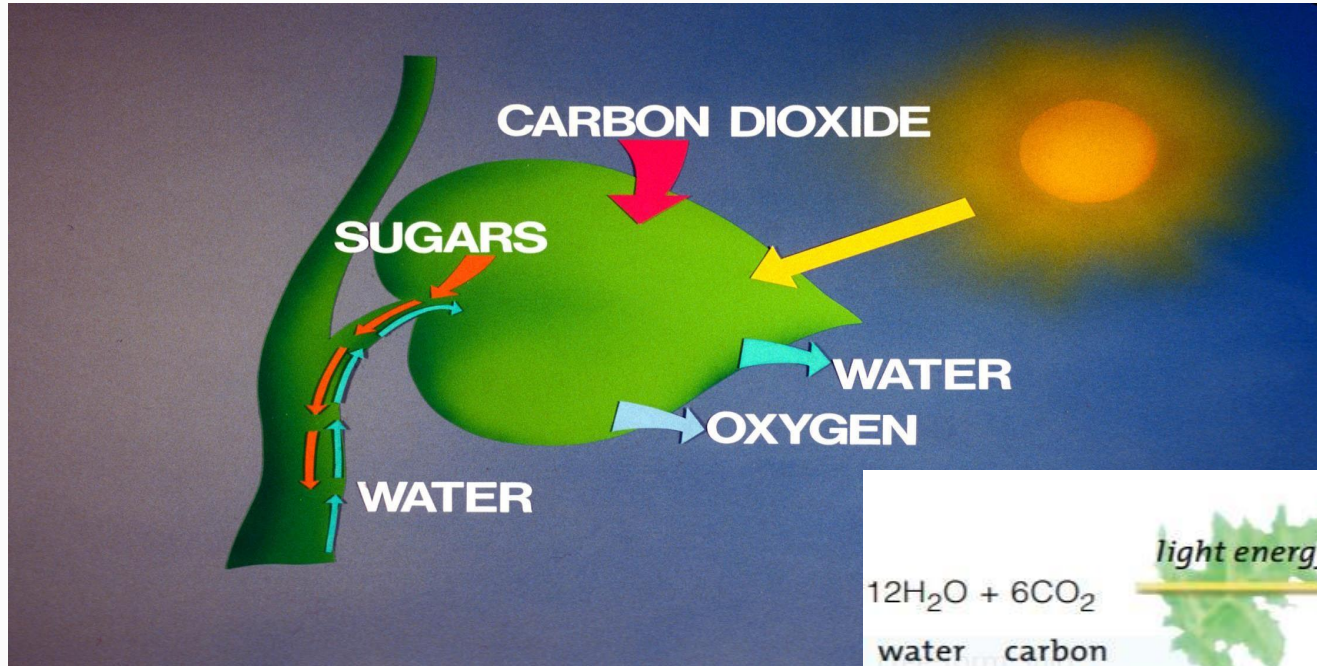


Euglena, a protist with both animal (movement by flagella, ingestion of small protozoans) and plant (carry out photosynthesis in its chloroplasts) characteristics



Cyanobacteria (also known as 'blue-green algae') are bacteria capable of photosynthesis- chlorophyll present in cytoplasm

Light Energy Harvested by Plants & Other Photosynthetic Autotrophs (Photoautotrophs)



Light

Light is made of photons, which are individual packets of electromagnetic energy traveling in waves. The shorter a photon's wavelength, the higher its energy.

For example, blue light has a shorter wavelength and more energy than red light.

Pigments

Pigments are a class of molecules that absorb photons with **particular wavelengths**. Photons that a pigment cannot absorb bounce off or continue on through it; they are reflected or transmitted.

Chlorophyll a is the most abundant type in plants, green algae, and a number of photoautotrophic bacteria.

Variety of Pigments

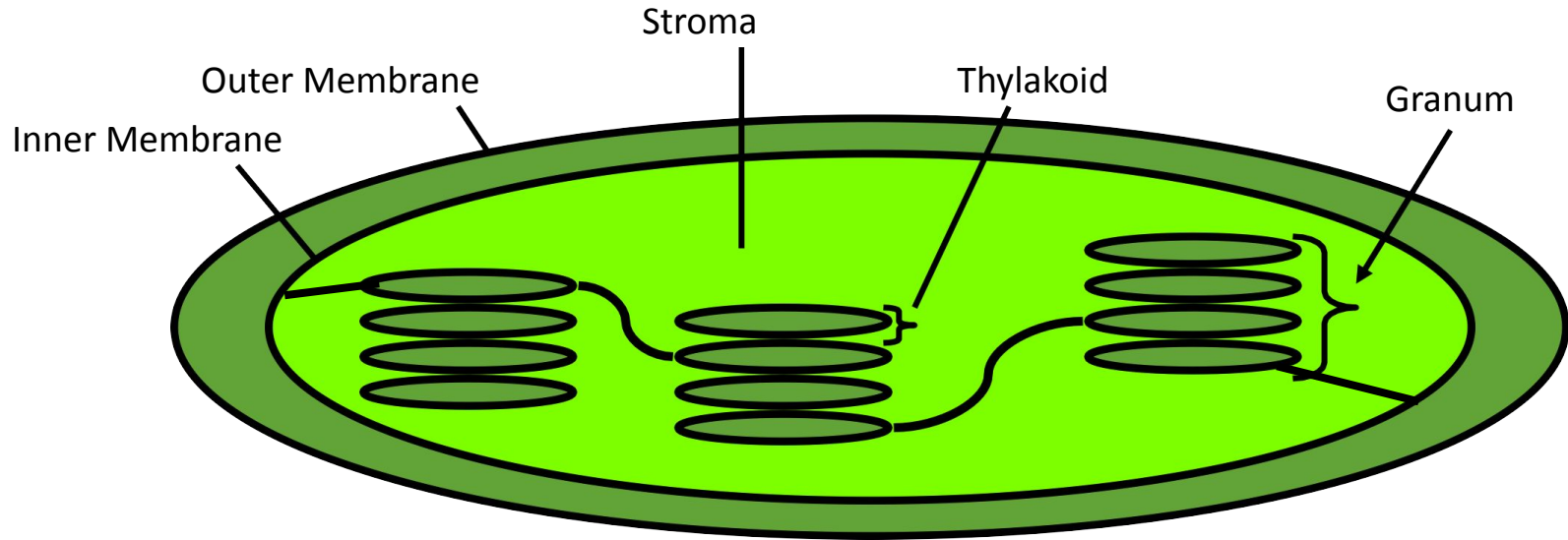
Pigment	Colour reflected	Found in
Chlorophyll a	Green-yellow	Plants, algae, cyanobacteria
Chlorophyll b	Green	Plants
Carotenoids	Yellow, orange, red	Plants, algae, bacteria, fruits, vegetables, flowers
Anthocyanins	Red, purple, blue	Plants, fruits, flowers
Phycobillins	Blue, green, yellow, brown, black, red	Algae

Color reflected is the color we see!

Where does photosynthesis take place?

Chloroplast!!!

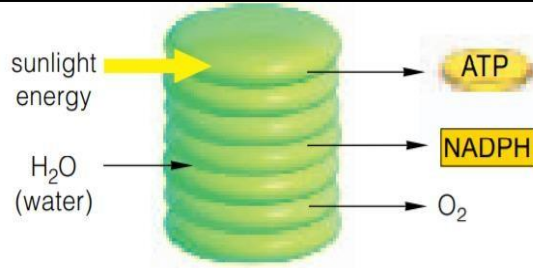
Organelle where **photosynthesis** takes place.



Thylakoid stacks are connected together.

How does photosynthesis take place?

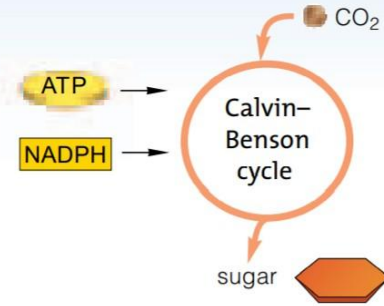
AN OVERVIEW OF PHOTOSYNTHESIS



Making ATP and NADPH

In the first stage of photosynthesis, sunlight energy becomes converted to chemical bond energy of ATP. Water molecules are broken apart, NADPH forms, and oxygen escapes into the air.

Light dependent stage



Making Sugars

The second stage is the "synthesis" part of photosynthesis. ATP delivers energy to reaction sites where sugars are built with atoms of hydrogen (delivered by NADPH), carbon, and oxygen (from carbon dioxide in the air).

Light independent stage/ Dark stage/ Calvin-Benson Cycle

AN OVERVIEW OF PHOTOSYNTHESIS

Photosynthesis occurs in two stages in chloroplasts. In the first stage, energy from the sun is converted to chemical energy and stored in ATP. NADPH is formed. These molecules are later used in the second stage (light independent reactions) to assemble sugars from carbon dioxide.

In the first stage, chloroplasts split H_2O into hydrogen and oxygen using light energy, incorporating the electrons of hydrogen into NADPH molecules and releasing oxygen as a by-product.

H_2O is the electron source!

Two Parts of Photosynthesis

Two stages make up photosynthesis:

1. Light Dependent Stage -> light dependent reactions

- Produces energy from solar power (photons) in the form of ATP. NADPH is formed.

- $\text{ADP} + \text{P} \rightarrow \text{ATP}$ (energy stored in pyrophosphate bond)
- $\text{NADP}^+ + 2\text{e}^- + \text{H}^+ \rightarrow \text{NADPH}$

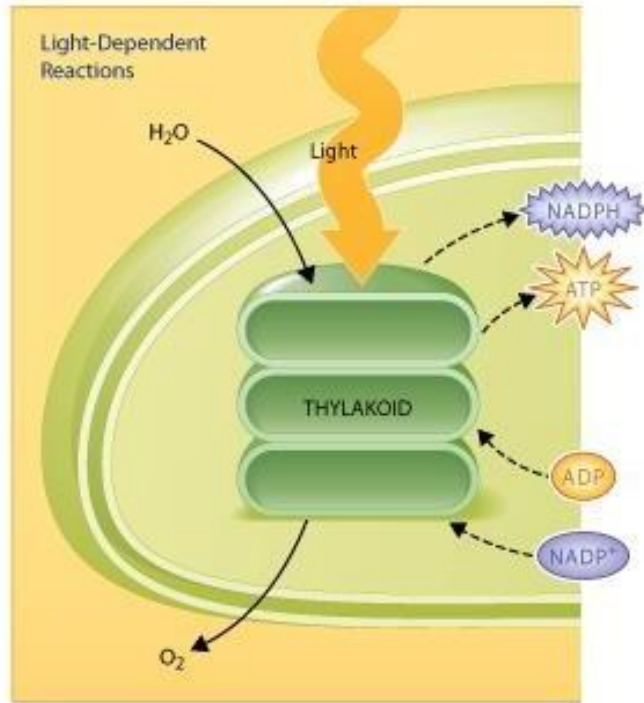
Two Parts of Photosynthesis

2. Light Independent Stage/ Dark Stage/ Calvin Cycle

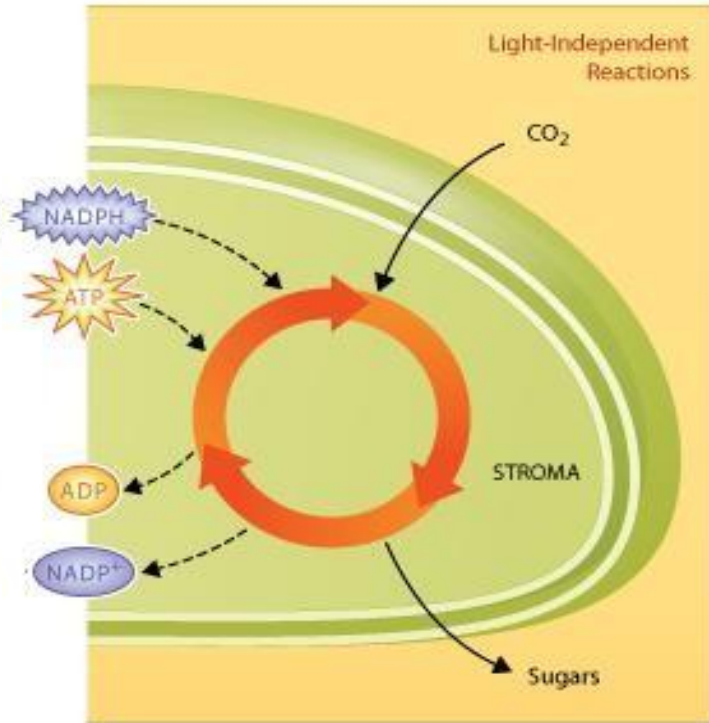
-> Light independent reactions

- Also called **Carbon Fixation** or **C₃ Cycle**
- Uses energy (**ATP**) and **NADPH** from light dependent stage reaction to make sugar (**glucose**) with **CO₂**.

Light vs. Dark Reactions



Copyright © Pearson Education, Inc., or its affiliates. All Rights Reserved.



Copyright © Pearson Education, Inc., or its affiliates. All Rights Reserved.

Remember!

- Photosynthesis creates glucose from carbon dioxide, water and light.
- Plants need glucose to store energy and to build cellulose.
- However, photosynthesis does not produce enough ATP for the cell's purposes.
- Plant cells also have mitochondria to produce ATP through **respiration**.

Are Plants Really That Important?

- Without plants, most of **life** on earth as we know it would not exist.
- Plants comprise about **98%** of the earth's biomass!
- Plants are primarily responsible for creating our **oxygen-rich** atmosphere via the **light reactions of photosynthesis**.
- Plants are Earth's main **autotrophs and fixers** of carbon.
- Plants provide **habitat, food and other necessities** upon which almost all other living organisms ultimately depend.
- Plants are involved in various **symbiotic relationships** with other organisms.

Thank you!!!

